

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN
(EVENING AND WEEKEND CLASSES)

SATURDAY, 14 MAY 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in Section A and **ONLY ONE (1)** question in Section B. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are /are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**Section A (Compulsory Questions)**

Q1. Suppose you are a system analyst working at sport complex. The current sport system installed at the sport complex only allows a customer to book a venue for sports' facilities. The project team leader has assigned you to design a membership loyalty module in order to improve the current sport system. This membership loyalty module enables the customer to earn loyalty points and to redeem gifts. You are required to use **throwaway prototyping** to develop this system. In addition, the project manager has created a work plan using **Gantt chart** for you.

Given the mentioned scenario, prepare the following items:

- (a) List **SIX (6)** factors that are used to select a methodology to develop the project. (6 marks)
- (b) Explain a situation where incomplete and inconsistent requirements specification to occur. (2 marks)
- (c)
 - (i) Suggest an appropriate requirement analysis technique to use in order to analyze the requirements in the project. (1 mark)
 - (ii) Discuss an activity that can be applied to select the requirement analysis technique suggested in Q1(c)(i). (3 marks)
- (d) Explain **THREE (3)** elements should be included in the Gantt chart. (3 marks)
- (e) Briefly discuss how you are going to use interview as a requirement gathering technique to collect the information from users. (10 marks)

[Total : 25 marks]

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- Q2. (a) The Age of Empire is an online game. A player is required to register his/her username, password, and email address to create an account. Later, the player is required to activate his/her game account via email. Upon activation, the player can control various villagers and military units. The primary function of the villagers unit is to collect resources, cut down trees for woods, mine for stones & gold and farm for food. The clubmen and swordsmen are the types of military units used to attack other player's civilizations. In addition, the player can use the resources to construct and upgrade buildings such as town center, house, and etc. in order to perform various operations. Other than these, the player can conduct research to upgrade the technology used in the military and villagers units.

Given the mentioned scenario, use the class diagram notation to draw the following items:

- (i) Generalization relationships between classes. (4 marks)
 - (ii) Aggregation relationships between classes. (2 marks)
 - (iii) Others association relationships between classes. (9 marks)
- [Hints: Combine the answer (i), (ii) and (iii) into a class diagram]
- (b) Explain **THREE (3)** types of connasence in terms of class design and method design. (6 marks)
- (c) Discuss the major deliverable for the first phase and second phase of the rational unified process. (4 marks)
- [Total : 25 marks]

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- Q3. (a) Bank X has hired you as a project manager to design and build an online queue ticketing system. During the queue number generating process, the customer is required to select the service(s) that they require via the queue ticketing system. Then, the customer will scan the QR code at the front door of the Bank X to get a queue number from the system. The customer can monitor the current queue number at each counter in real-time. In addition, the system will alert the customer upon his/her turn to go to the counter. On the other hand, each staff in the Bank X can generate a queue number to receive the customers to their respective counter. However, each counter only handles specific services. In this case, the system will not allow the customer to attend to the counter(s) that does not provide the required service(s).

Given the mentioned scenario, use the use case diagram notation to draw the following items:

- (i) Actors who will interact with the system. (2 marks)
 - (ii) Use case that associated with actor(s). (5 marks)
 - (iii) Use case that associated with others use case(s). (4 marks)
 - (iv) System boundary. (1 mark)
- [Hints: Combine the answer (i), (ii), (iii), and (iv) into a use case diagram]
- (b) Explain the differences between functional model and behavioral model in term of unified modeling language (UML). (4 marks)
 - (c) Explain the concept of “generalization” and “specialization” in term of the class diagram relationship. Illustrate an example to explain both concepts. (7 marks)
 - (d) List the components of the activity diagram that are used to model a parallel process. (2 marks)
- [Total : 25 marks]

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**Section B (Choose any ONE Question)**

Q4. (a) Given the flow of events:

- i. The citizen selects a type of public services such as road issue, power supply issue, water supply issue, intan water issue and etc. to launch a complaint.
- ii. The citizen enters the particulars of the complaint.
- iii. The citizen submits the complaint.
- iv. The system generates a reference number and sends the submitted complaint to the administration staff to resolve the issue.
- v. The administration staff appoints in-house technical staff to resolve the complaint.
- vi. If the in-house technical staff cannot resolve the issue, the administration staff appoints a vendor to resolve the complaint.
- vii. If the vendor cannot resolve the issue, the administration staff appoints another vendor to resolve the complaint. This step will be repeated until the complaint has been resolved.
- viii. The administration staff notifies the citizen that the complaint has been resolved.
- ix. The system sends a survey to the citizen for rating.
- x. The citizen responds to the survey.
- xi. The system saves the feedbacks and rating by the citizen.

Given the mentioned scenario, use an activity diagram with swimlane notation to illustrate the following items:

- (i) Swimlane partitions. (3 marks)
- (ii) Step-by-step activities for complaint launching process. (5 marks)
- (iii) Decision node and iteration used for complaint launching process. (5 marks)
- (iv) Usage of the initial node, final activity node, merge node and control flow notation. (2 marks)

[Hints: Combine the answer (i), (ii), (iii) and (iv) into an activity diagram with swimlane]

- (b) Discuss **THREE (3)** types of messages used to model the message passing among the objects in the sequence diagram. (6 marks)
- (c) Explain the differences between two-tier and three-tier architecture in term of system architecture design. (4 marks)

[Total : 25 marks]

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- Q5. (a) Explain **FIVE (5)** major steps of a user interface design process in design phase.
(10 marks)
- (b) Discuss **THREE (3)** major roles of a project manager to manage programming during implementation phase.
(6 marks)
- (c) Suggest and explain **ONE (1)** type of design strategy used to develop a system in design phase.
(3 marks)
- (d) Identify **FOUR (4)** types of the operational requirements used in design phase.
(4 marks)
- (e) List **TWO (2)** types of the object persistent format used in data storage design.
(2 marks)
- [Total : 25 marks]
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